
Huijie Yang

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EDUCATION

UT Health Science Center San Antonio

Ph.D. student in Integrated Biomedical Sciences
GPA (Present): 4.0/4.0

08/2024-Present

Central South University, Second Xiangya Hospital

Master of Medicine (Internal Medicine)
GPA:3.30/4.0

09/2020-07/2023

University of South China

Bachelor of Medicine(Clinical Medicine)
GPA:3.78/4.0

09/2015-06/2020

RESEARCH INTERESTS

- ◆ Computational biology; multi-omics data integration; single-cell transcriptomics
- ◆ Genetic variation; three-dimensional chromatin organization; epigenomics
- ◆ Cancer genomics; hepatocellular carcinoma; aging; pancreatitis

RESEARCH EXPERIENCES

Integrative multi-omics analysis of epigenetic and 3D genome regulation in MET- and MYC-driven hepatocellular carcinoma

Role: Project leader, PI: Dr. Yu Luan

11/2025-Present

- ◆ I designed the research strategy for the project. Mapped the regulatory architecture of the tumor microenvironment by integrating Hi-C, RNA-seq, and ATAC-seq data in MET- and MYC-driven HCC mouse models.
- ◆ I conducted a comprehensive assessment of the 3D genomic landscape (A/B compartments, topologically associated domains, and chromatin loops), integrating differentially accessible regions and differentially expressed genes. Revealed that the global downregulation of complement and interferon-related genes in MYC-driven tumors is directly orchestrated by 3D chromatin conformational changes.
- ◆ Model-specific TF regulatory networks were constructed using PECA (paired expression and chromatin accessibility) to identify key TFs, such as Lef1 and Tcf3, as potential novel therapeutic targets for reversing immunosuppressive states.

Characterizing rapamycin-responsive myeloid population in aging

Role: Project leader, PI: Dr. Yu Luan, collaborate with Dr. Ellen Kraig

05/2025-Present

- ◆ Performed scRNA-seq analysis on CD11b⁺ PBMC cells (P2 and P3 cell populations) to determine the transcriptional characteristics of myeloid cells that respond to the rapamycin.
- ◆ Weighted gene co-expression network analysis (WGCNA) was performed on CD14⁺ monocytes, identifying 55 gene features (such as NRG1) that are highly enriched in the P3 cell population.
- ◆ I generated critical preliminary single-cell data and the analysis framework, supporting the writing and submission of the NIH R21 grant application.

Identification and functional characterization of pancreatitis risk variants

Role: Major Researcher, PI: Dr. Yu Luan, collaborate with Dr. Pei Wang

02/2025-Present

- ◆ Collected 204 pancreatitis-risk SNPs from 12 GWAS studies and screened ~2M regulatory-region candidates.
- ◆ Integrated with ATAC-seq, ChIP-seq, and Hi-C data to prioritize functional SNPs associated with pancreatitis risk, revealing that rs545634 creates a novel KLF9 binding motif near the pancreas-specific gene CTSC, while rs125959755 disrupts a CTCFL motif at a TAD boundary and alters long-range chromatin interactions with pancreas-related genes such as CELA2B.
- ◆ Bridged the gap between statistical genetic risk and molecular mechanisms, revealing how specific non-coding variants drive pancreatitis pathogenesis through 3D chromatin interactions.

Single-cell RNA sequencing reveals tumor progression and microenvironment features in immune-hot and immune-cold of hepatocellular carcinoma (manuscript under preparation)

Role: Project leader, PI: Dr. Yu Luan, collaborate with Dr. Luzhe Sun

10/2024-Present

- ◆ Integrate single-cell and bulk RNA sequencing data from MET- and MYC-driven HCC mouse models at different stages (2, 5, and 10 weeks) to depict the temporal dynamics of the tumor microenvironment.
- ◆ Uncovered divergent immune trajectories during tumor progression, demonstrating that MET-driven tumors maintain an actively inflamed state, whereas MYC-driven tumors rapidly transition to an immune-cold state.
- ◆ The MYC-specific transcriptional program driving T cell exclusion provides a molecular framework for the immune-cold phenotype, identifying SOX4 and HK2 as potential markers linking oncogenic signaling to immune evasion and poor prognosis.

Biological mechanisms of Chlamydia-gut interactions

Role: Major Researcher, PI: Dr. Guangming Zhong

09/2023-08/2024

- ◆ Performed intrajejunal CMpGP3S inoculation in knockout mice and quantified colonization in tissues/rectal swabs.
- ◆ Characterized the specific intestinal immune cell populations and dynamic responses induced by the invasive CMpGP3S strain, utilizing immunofluorescence (IF) and flow cytometry.
- ◆ Elucidated the distinct immune mechanisms driving CMpGP3S dissemination from the small intestine to the colon and the direct colonic colonization of wild-type Chlamydia, revealing how host intestinal immunity actively exerts selective pressure to shape gut microbiota composition.

Mechanism of immune cells involved in the pathogenesis of diabetic cardiomyopathy

Role: Major Researcher, PI: Dr. Yang Xiao

05/2021-05/2023

- ◆ Recruited clinical participants, collected left ventricular myocardial tissue from patients during surgery, and involved in the analysis of snRNA-seq of myocardial samples.
- ◆ I conducted experiments to validate the sequencing results, revealing the role of communication between macrophages and fibroblasts in promoting myocardial fibrosis in diabetic cardiomyopathy.
- ◆ Targeting PDGFC may serve as a novel therapeutic approach for treating diabetic cardiomyopathy.

Intestinal Microbiota and Pathogenesis of Hypertension

Role: Project leader, PI: Dr. Zhuowei Hu

10/2020-06/2021

- ◆ Designed and conducted experiments exploring the interaction between gut microbiota and hypertension, focusing on the combined therapeutic effects of berberine and stachyose.
- ◆ Proficient in hypertension model surgery, detection of bacterial species expression in feces, and blood pressure measurement in rats and mice. The combined treatment of berberine and stachyose is more effective than monotherapy.
- ◆ Revealed that improving the gut barrier has a certain beneficial effect on hypertension but is insufficient to reverse severe hypertension phenotypes.

PUBLICATIONS

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- ◆ **Yang, H. (Co-first author)**, Zhang, W., Ding, J., Hu, J., Sun, Y., Peng, W., Chu, Y., Xie, L., Mei, Z., Shao, Z., Xiao, Y. A novel genomic instability-derived lncRNA signature to predict prognosis and immune characteristics of pancreatic ductal adenocarcinoma. *Frontiers in immunology* 13, 970588, doi:10.3389/fimmu.2022.970588 (2022).
 - ◆ Xie, L., Xiao, Y., Tai, S., **Yang, H.**, Zhou, S., Zhou, Z. Emerging Roles of Sodium Glucose Cotransporter 2 (SGLT-2) Inhibitors in Diabetic Cardiovascular Diseases: Focusing on Immunity, Inflammation and Metabolism. *Frontiers in pharmacology* 13, 836849, doi:10.3389/fphar.2022.836849.
 - ◆ Xu, Y., Wang, Y., Winner, H., **Yang, H.**, He, R., Wang, J., Zhong, G. Regulation of chlamydial spreading from the small intestine to the large intestine by IL-22-producing CD4⁺ T cells. *Infect Immun.* 2024;92(1):e0042123. doi:10.1128/iai.00421-23

ORAL & POSTER PRESENTATIONS

◆ <u>2026 Mays Cancer Center Joint T32 Retreat</u> , Poster	05/2026
◆ <u>2026 CSA Annual retreat</u> , Poster	04/2026
◆ <u>Mikiten Research Symposium</u> , Poster and 3MT Competition	02/2026
◆ <u>2026 Mays Cancer Center Research Symposium</u> , Poster	02/2026
◆ <u>2025 San Antonio Liver Cancer Symposium</u> , Poster	10/2025
◆ <u>2025 CSA Annual retreat</u> , Poster	04/2025
◆ <u>2025 Mays Cancer Center Research Symposium</u> , Poster	02/2025

AWARDS

Honorable Mention, PhD Student Poster Category, Mays Cancer Center Research Symposium, San Antonio, TX. 2026

The Exceptional Poster Presentation Award, 2026 Mays Cancer Center Joint T32 Retreat, San Antonio, TX. 2026

David Carrillo Memorial Fellowship Award, 2026 Cell System & Anstomy Retreat, San Antonio, TX. 2026

CLINICAL EXPERIENCES

Nanhua Hospital Affiliated with University of South China

Role: Intern 03/2018-12/2018

The First People's Hospital of Huaihua

Role: Intern 01/2019-12/2019

- ◆ Obtained Certificate of Licensed Practising Physician in China in 2021.

SKILLS

- ◆ **Bioinformatics & Data Analysis:** scRNA-seq, bulk RNA-seq, ATAC-seq, ChIP-seq, Hi-C
 - ◆ **Programming Languages:** R, Python (basic)
 - ◆ **Experimental skills:** Animal handling (gavage, injections, blood pressure/glucose measurement), cell isolation and culture (islets, immune cells, fibroblasts, DCs), molecular assays (flow cytometry, IF/IHC, qPCR, WB, ELISA), and chlamydia infection models in mice.
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